NeMo-map: Neural Implicit Flow Fields for Spatio-Temporal Motion Mapping

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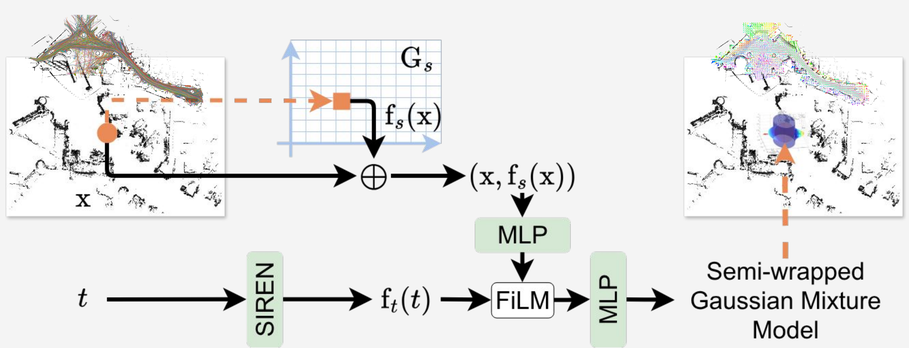
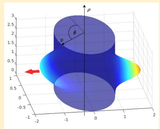
Introduction

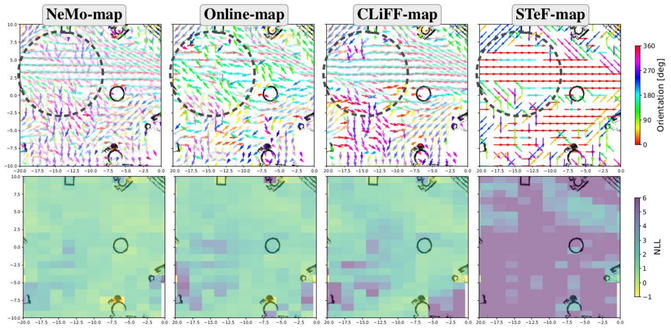
- Propose **NeMo-map**: A continuous spatio-temporal *Map of Dynamics* using implicit neural representations, achieving more **accurate** and **efficient** human motion pattern representation.
- Maps of dynamics (MoDs) encode motion patterns that are:
  - multimodal
  - structured by geometry & physics
  - change over space and time
- Evaluated on two real-world datasets, showing:
  - better accuracy of motion representation
  - practicality and efficiency
  - improved prediction accuracy in the downstream task: long-term human motion prediction



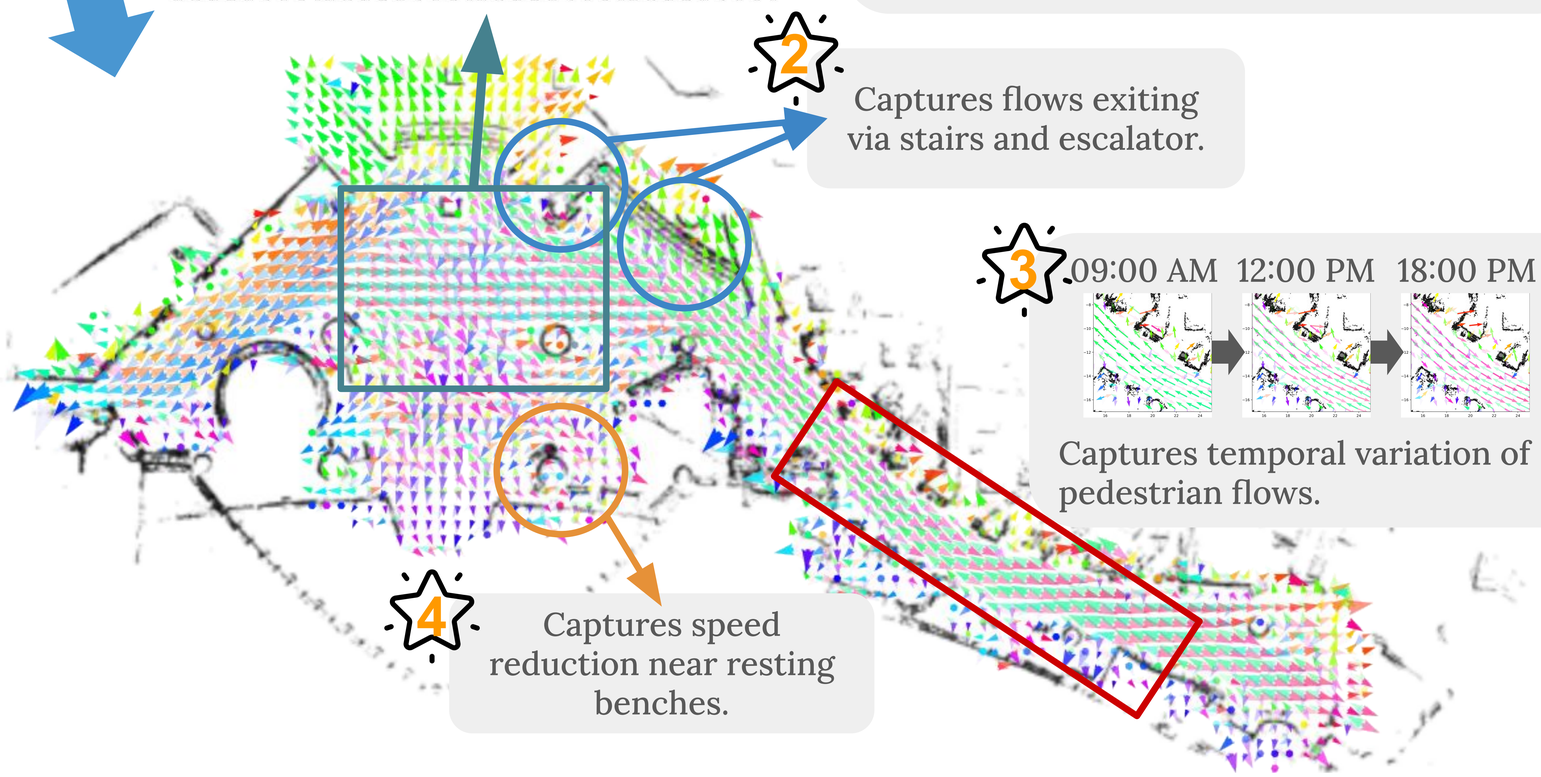
NeMo-map: Learn a **continuous** motion field that maps

(x, y, t) → 

Spatio-temporal coordinate → Motion distribution



1 Preserves both motion modes where horizontal and vertical flows intersect, while baseline MoDs break the continuity of vertical flow.



2 Captures flows exiting via stairs and escalator.

3 09:00 AM 12:00 PM 18:00 PM

Captures temporal variation of pedestrian flows.

4 Captures speed reduction near resting benches.